Roberto Villafuerte

AI Engineer | Quito, Ecuador **contact@villafortech.com**

I design and build AI and data systems, with an emphasis on the architecture that makes them reliable, secure, and maintainable.

Portfolio villafortech.com
LinkedIn linkedin.com/in/robertovillafuerte
GitHub github.com/VillaforTech

EDUCATION

01 **BEng, Computer Science Engineering**
Universidad San Francisco de Quito (USFQ)
2022 - 2026

Scholarship recipient. Final semester with capstone project in progress.

02 **Exchange studies, Data Science**
University of Helsinki
August - December 2025

Average: 4.83/5.0. MSc-level courses: Engineering of ML Systems, Big Data Platforms, Trustworthy ML, and Software Architectures.

LEADERSHIP & TEACHING

Coordinator
AI Safety Ecuador 2025

Competitive Programming Instructor
Competitive Programming Club 2025

CREDENTIALS

Technical AI Safety
BlueDot Impact January 2026

Machine Learning Specialization
IBM October 2025

2nd Place, IEEEXtreme 18.0, Ecuador
IEEE October 2024

LANGUAGES

English	Native or bilingual proficiency
Spanish	Native or bilingual proficiency
Chinese	HSK 2

EXPERIENCE

01 **Applied AI Engineer** August 2026 - Present
COMPUMAX

- Translate product and business requirements into architecture, data flows, backend services, and implementation plans.
- Develop AI and data product capabilities with reliability, security, maintainability, and scalability in view.

02 **Lead Data Architect** February 2026 - June 2026
SaliHub

- Designed PostgreSQL structures and migrations for a multi-tenant health platform, including access controls, consent, auditability, and reporting.
- Added database health checks, initial tests, and source-aware deduplication across devices; documented technical decisions.

03 **AI Engineer** December 2025 - February 2026
COMPUMAX

- Built a support-automation system with FastAPI, Chatwoot, Qdrant, PostgreSQL/pgvector, Redis, and Docker Compose.
- Implemented retrieval, intent routing, human escalation, logging, and internal adapters, with a human-only fallback.

04 **Machine Learning Engineer Intern** February 2025 - June 2025
Datalysis Group

- Developed Python and Mage pipelines for data preparation, feature engineering, model evaluation, prediction, alerts, and export.
- Integrated PostgreSQL ingestion and export, kept predictions separate from evaluation metrics, and corrected schemas, write policies, and documentation.

SELECTED PROJECTS

Auditable ML policy lab
Academic project · University of Helsinki

Classification lab with separate training, validation, and test stages; fairness reports, traceable artifacts, and an API with explicit contracts.

Case study and code

Bio-inspired optimization benchmark
Open-source project · Python

Reproducible comparison of GA, PSO, ACO, and GWO on test functions, routes, and schedules, with configurable seeds, saved metrics, and convergence curves.

Code on GitHub

SELECTED SKILLS

Data Architecture · Machine Learning Engineering · Data Engineering · Software Architecture · Database Design · Machine Learning · Generative AI · MLOps · PostgreSQL · Database Security · Backend Development · Cloud Computing · Python · SQL · Docker